Competition 03 — AI and Data Science Hackathon

Overview

The AI and Data Science Hackathon challenges participants to work with a real-world dataset, extract meaningful insights, and build a data-driven solution under time pressure. There are no presentations — work is evaluated entirely on the quality of the submitted notebook and analysis.

Day 1 is an open analysis round on an accessible dataset. A select number of top performers are shortlisted and invited to Day 2, where they receive a harder dataset with a shorter time window and compete again for the final ranking. The module management reserves the right to determine how many participants qualify for Day 2, and not all submissions will advance.

Participation format: Solo, Duo, or Triplet

Venue: DHA Suffa University

Duration: 2-Day Event (June 8 and 9, 2026)

Tools allowed: Python with Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn. Jupyter Notebook or Google Colab. TensorFlow and PyTorch are optional. Power BI or Tableau for visualisation. Any public pre-trained model or API. R is optional. No web app building required. A well-documented Jupyter notebook with all cells run and outputs visible is a complete submission.

Registration Fees

Solo — Rs. 500

Duo — Rs. 1,000

Triplet — Rs. 1,500

Timeline and Format

Pre-event — Domain Announcement (approximately one week before June 8)

A broad domain is announced ahead of the event — for example healthcare, agriculture, education, or urban mobility. This allows participants to read background literature and think about the problem space. The actual dataset is not released until the morning of Day 1.

Day 1 — Explore and Model (June 8)

Dataset distributed at 9 AM. Participants work through the full data science pipeline: data understanding, cleaning, exploratory data analysis, feature engineering, modelling, evaluation, and insight generation.

By the Day 1 submission deadline, each participant or team must submit a final Jupyter notebook with all cells run and outputs visible, and a one-paragraph summary covering what problem they tackled, what they found, and why it matters.

Submissions are reviewed overnight by the internal evaluation panel. A select number of top performers are invited to Day 2. The exact number of qualifiers is determined by the module management. Not all Day 1 submissions will advance.

Day 2 — Final Round (June 9)

Finalists receive a harder dataset with a shorter time limit. The same format repeats — work through the pipeline, submit the notebook by the deadline. No presentations. Evaluation is done by the panel after submissions close.

Winners and runner-up are decided based on Day 2 performance.

Results and Awards Ceremony

Winners announced at the closing ceremony. All top three are invited on stage.

Competition Tracks

Participants declare their track at the time of Day 1 submission. Both tracks use the same dataset.

Track A — Predictive Modelling Build a supervised learning model on the provided dataset. Focus is on strong, well-evaluated predictive performance with clear justification of all choices. Judged on methodology, evaluation rigour, and insight into model behaviour.

Track B — Exploratory Analysis and Visualisation Produce a deep exploratory analysis of the dataset — patterns, anomalies, correlations, and actionable insights — without necessarily building a predictive model. Judged on depth of analysis, quality of visualisations, and strength of narrative.

Judging Rubric

Problem understanding and EDA — 30%

The problem is clearly defined with a specific target variable or research question. Exploratory analysis is thorough — distributions, correlations, and outliers are examined. Data quality issues are identified and addressed with clear reasoning.

Methodology and modelling — 25%

The choice of algorithm or approach is justified, not arbitrary. Train/test split, cross-validation, or an equivalent is used correctly. Evaluation metrics are appropriate for the problem type.

Quality of insights — 25%

Results are interpreted in plain language, not just numbers. The team explains what the findings mean for the real-world problem. Limitations and failure cases are acknowledged honestly.

Notebook quality and reproducibility — 20%

The notebook is clean, well-commented, and fully re-runnable from top to bottom. Visualisations are clear and support the story being told. Use of any pre-trained models or external APIs is clearly cited.

High accuracy alone is not sufficient. Judges value honest analysis and genuine insight over optimised leaderboard numbers.

Prize Breakdown

1st place — Rs. 20,000

2nd place — Rs. 10,000

3rd place — Certificate and goodies, recognised on stage at the awards ceremony

Total prize pool: Rs. 30,000

Rules and Regulations

Solo, duo, and triplet participation is allowed.

The dataset is distributed at 9 AM on Day 1. No prior access is permitted under any circumstances.

All analysis must be performed during official event hours. Pre-analysed work is grounds for disqualification.

Any Python, R, or BI tool is permitted. Use of public pre-trained models and AI APIs is allowed — cite them clearly in the notebook.

The notebook must be fully re-runnable from top to bottom with all outputs visible at the time of submission.

Teams may not share code, notebooks, or analytical approaches with other registered participants.

Any violation results in disqualification with no refund.